

Converting a quantitative endpoint profile into a face bound

Research lemma: independently reviewed

30 September 2026

This is a conditional quantitative strengthening of the fully reviewed three-halves argument in `../power_per_die_lower_bound.tex`. It does not assert the additional profile hypothesis without proof.

Let $p_j : [K] \rightarrow [0, 1]$ be nondecreasing, with positive row weights w , and suppose $\mathbb{E}_w \prod_{j \in S} p_j = 1/(|S| + 1)$ for every $S \subseteq [m]$. Put $u_j = 1 - p_j$, $s = \sum_j u_j$, $\ell = \lfloor m/4 \rfloor$, $\alpha = \ell/m$, and

$$H = \ell(\max_j u_j - \min_j u_j).$$

The finite endpoint-balance theorem gives $H \leq C(s + 1)$ and

$$\mathbb{E}_w (s + 1)^k e^{-as} \leq C_a^{k+1} k! / m \quad (k \geq 0) \quad (1)$$

for every fixed $a > 0$.

Theorem 1. *There are absolute constants $c, C > 0$ with the following property. Suppose, for some $1 \leq L \leq m/256$ and $0 \leq \delta \leq 1$, that*

$$H \leq \delta(s + 1) \quad \text{on all rows with } s \leq L. \quad (2)$$

Write $\rho = \delta + e^{-cL}$. Then

$$w_K \leq C \max\{m^{-2}, m^{-3/2} \sqrt{\rho}\}.$$

For an equal-weight representation,

$$K \geq c \min\{m^2, m^{3/2} / \sqrt{\rho}\}.$$

The analogous estimate for w_1 holds if the corresponding reflected profile hypothesis is available.

Proof. Retain $\theta = 1/16$, the success-subset mixture $\mu = (\ell + 1)w\bar{A}$, $A_B = \prod_{j \in B} p_j$, and the beta law β_ℓ from the three-halves proof. For Laguerre sums $Q_a(z) = \sum_{j=0}^d c_{aj} L_j(\theta z)$ write $B_a = \sum |c_{aj}|$ and $B = B_1 B_2$. We first prove, for an absolute sufficiently small $c_0 > 0$ and $1 \leq d \leq c_0 m$, the uniform estimate

$$|\mathbb{E}_\mu Q_1(\alpha s) Q_2(\alpha s) - \beta_\ell(Q_1 Q_2)| \leq CB \frac{d}{m} (\delta + e^{-cL}). \quad (3)$$

Taking $c_0 < 3/8$ ensures that the degree is below the number of remaining columns.

For a complement R of size $N = m - \ell$, set $z_j = \ell u_j$ and $v = N^{-1} \sum_{j \in R} z_j$. The exact polarization expansion, the entire centered-coefficient estimate, and the variance-adapted Laguerre derivative bound give a pointwise error at most

$$B e^{\theta v} \sum_{k=2}^{2d} \frac{(C \sqrt{dkH/m})^k}{k!}.$$

Indeed $\sum_{j \in R} (z_j - v)^2 \leq NHv$, so all singular powers of v cancel. If $v = 0$, the error is identically zero. On $s \leq L$, substitute $H \leq \delta(s + 1)$, multiply by the success weight, and use (1). The resulting sum is bounded by

$$CB \sum_{k=2}^{2d} (C \sqrt{d\delta/m})^k (k + 1) \leq CB d \delta / m.$$

On $s > L$, reserve half of the exponential decay before applying (1): $e^{-as} \mathbf{1}_{s > L} \leq e^{-aL/2} e^{-as/2}$. The coarse bound $H \leq C(s + 1)$ then gives

$$CB e^{-cL} \sum_{k=2}^{2d} (C \sqrt{d/m})^k (k + 1) \leq CB (d/m) e^{-cL}.$$

Both geometric estimates hold by choosing c_0 absolutely small.

For completeness the success-subset averaging admits the same gain. On $s \leq m/256$ all $p_j \geq 1/2$. For $x = \alpha s$ and $\Delta = v - x$, under the success-weighted subset law the exact bias identity and negative inclusion covariances give

$$0 \leq \mathbb{E} \Delta \leq CHx/m, \quad \text{Var } \Delta \leq CHx/m.$$

For the variance, subtract $\min_j z_j$ from all coefficients; the subset size is fixed, and the remaining nonnegative coefficients have squared sum at most $H \sum_j z_j$. The singular Taylor estimate for $P = Q_1 Q_2$ is

$$|P(v) - P(x) - P'(x)\Delta| \leq CB d e^{\theta \max(x,v)} \Delta^2 / x, \quad |P'(x)| \leq CB \sqrt{d/x} e^{\theta x}.$$

At $x = 0$ all deficits vanish. Thus, after multiplication by $\bar{A} \leq e^{-\alpha s}$ and normalization, the error is at most

$$CB \frac{d}{m} m \mathbb{E}_w \left[(H(1+s) + H^2(1+s)/m) e^{-\alpha s} \right].$$

Split at L , use (2) and (1) on the lower part, and reserve half the exponential on the upper part. The resulting bound is $CB(d/m)(\delta + e^{-cL})$. The remaining rows $s > m/256$ have error $CB \text{poly}(m) e^{-c_1 m}$ by the crude uniform-subset comparison. Decreasing c absorbs this into the stated bound, since $L \leq m/256$ and $d \geq 1$. This proves (3).

The same Laguerre taper has a positive beta expectation bounded below absolutely, and factors into Laguerre sums with coefficient norms $O(d)$ and $O(1)$. Hence its comparison error is $Cd^2\rho/m$. Choose

$$d = \left\lfloor c_2 \min\{m, \sqrt{m/\rho}\} \right\rfloor$$

with a sufficiently small absolute $c_2 > 0$. The taper then forces $x_K < 2/(\theta d)$. The normalized Christoffel square at this point has coefficient norm $O(1)$ and beta expectation $O(1/d)$, whence

$$\mu(K) \leq C/d + Cd\rho/m \leq C/d.$$

As $s_K = O(1/d)$, every $A_B(K) \geq 1/2$ for large m . Therefore $w_K \leq C/(md)$, exactly the claimed bound. $\square$

For example, a proved profile estimate with $L = A \log m$ and $\delta \leq m^{-\gamma}(\log m)^b$, where $0 < \gamma < 1$ and A is sufficiently large, would imply

$$K \geq c m^{3/2+\gamma/2} (\log m)^{-b/2}.$$

This last sentence is an implication, not an assertion that such a quantitative profile has already been established.